GenomicSuperSignature: Interpretation of RNA-seq experiments through robust, efficient comparison to public databases

CUNY SPH GRADUATE SCHOOL OF PUBLIC HEALTH & HEALTH POLICY

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Major Applications

- Compare a new dataset to large public databases (SRA, GEO, MSigDB, etc.)
- Compare heterogeneous datasets (e.g. disease, cell types, platforms, etc.)
- Identify RNA expression scores of biological phenomenon (e.g. disease subtypes, cell types, etc.)
- Understand and annotate weak or indirectly measured biological attributes

Purpose

Thousands of RNA sequencing profiles have been deposited in public archives, yet remain unused for the interpretation of most newly performed experiments. Methods for leveraging these public resources have focused on the interpretation of existing data, or analysis of new datasets independently, but do not facilitate direct comparison of new to existing experiments. The interpretability of common unsupervised analysis methods such as principal component analysis (PCA) would be enhanced by efficient comparison of the results to previously published datasets.

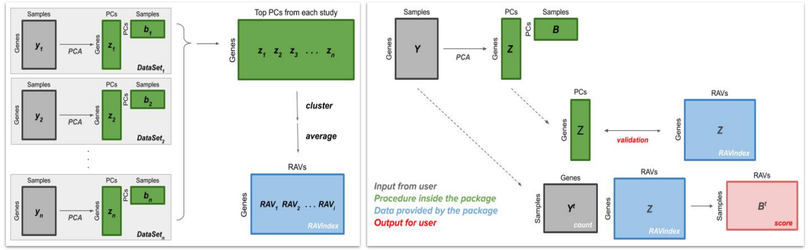
Methods

To help identify replicable and interpretable axes of variation in any given gene expression dataset, we performed PCA on 536 studies comprising 44,890 RNA sequencing profiles[1]. Sufficiently similar loading vectors, when compared across studies, were combined through simple averaging. We annotated the collection of resulting average loading vectors, which we call ***Replicable Axes of Variation (RAV)***, with details from the originating studies and gene set enrichment analysis. Functions to match PCA of new datasets to RAVs from existing studies, extract interpretable annotations, and provide intuitive visualization, are implemented as the **GenomicSuperSignature R package**, to be submitted to Bioconductor.

Results

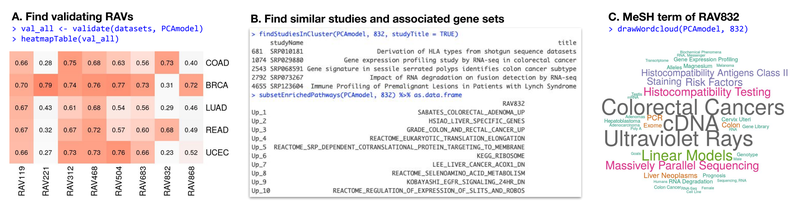
The process to generate RAVs is robust to the batch effects and the presence of low-quality or irrelevant studies. We demonstrate that the GenomicSuperSignature package allows nearly instantaneous matching of PCA axes in new datasets to pre-computed RAVs from hundreds of previous studies and thousands of samples on an ordinary laptop. GenomicSuperSignature thereby links, in negligible compute time, relevant studies from the public literature, and provides interpretation of potentially subtle biological signals through gene set enrichment analysis of loadings and MeSH term analysis of matching studies. We demonstrate RAVs associated with phenotype can provide insight into weak or indirectly measured biological attributes in a new study by leveraging accumulated data from published datasets. The performance of GenomicSuperSignature is compared to complementary previous works, and demonstrates 1) ability to identify replicable PCA loadings for colorectal carcinoma using a large, non-specific RNA-seq database instead of a focused microarray database, and 2) estimate neutrophil counts through transfer learning on new data comparably to the previous efforts despite major differences in training datasets and model building processes.

Implementation and application



Search existing databases using RAVs

From the expression data input, GenomicSuperSignature matches Principal Components to the existing database within seconds. Blue command lines are used for the below outputs. **A)** Five TCGA datasets are compared. **B)** Studies and GSEA results associated with RAV832, which is associated with colon and rectal cancer based on A). **C)** MeSH terms associated with RAV832.

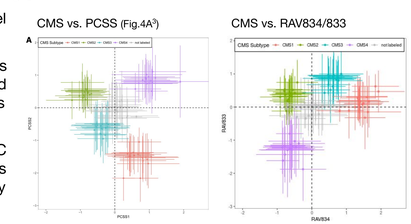


RAVs to characterise cancer subtypes

Discrete subtypes can fail to capture the cellular-level evolution or mislabel the mixed population of cells.

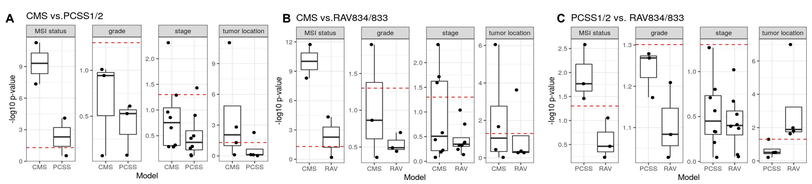
18 CRC datasets are plotted based on the continuous scores assigned by different scoring models, PCSS[3] and RAV, and colored based on the discrete CRC subtypes (CMS) proposed by the Consortium.

In spite of the major difference in training datasets, 8 CRC microarray datasets for PCSS versus 536 heterogeneous RNA sequencing datasets for RAV, we observed very similar performance on CRC subtyping.

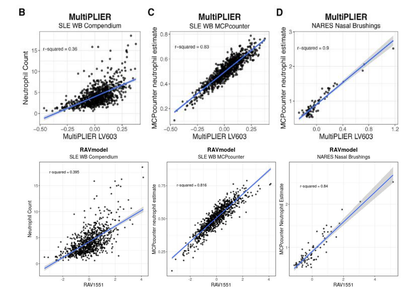


Generic RAVs vs. bespoke cancer-specific analysis

RAVs characterise CRC transcriptome variation comparably to bespoke continuous scores and better than published discrete subtypes. Likelihood-ratio test between the reference full model vs either continuous scores only, or discrete subtypes only, are plotted. A p-value near 1 ($-\log_{10}$p-value near 0) suggests that *no additional information is provided by the full model*. **A)** Performance of PCSS against CMS. **B)** RAV834/833 identified based on the metadata and validation. **C)** Comparison between PCSS and RAV. RAV834/833 are more closely associated with biological variables than PCSS1/2 except tumor location.



Project biological signal across datasets



RAV associated with biological signal can be applied across different datasets with comparable performance to the existing tool[2,4].

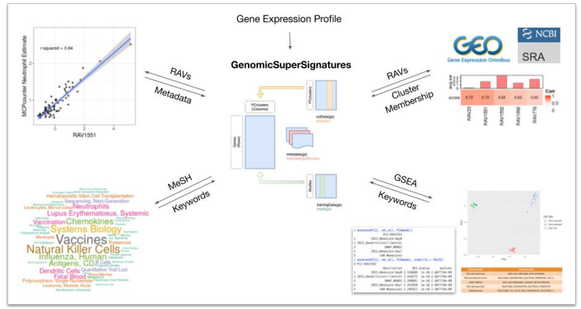
[Column 1-2] Neutrophil count of Systemic lupus erythematosus whole blood (SLE-WB) Compendium is from metadata (column 1) or estimated by MCPcounter[5] (column 2).
[Column 3] Neutrophil estimate of nasal brushings microarray dataset (NARES) by MCPcounter.

[Top] Neutrophil-associated LV603 from multiPLIER (= recount2 based PLIER model)[4]
[Bottom] Neutrophil-associated RAV1551 from GenomicSuperSignature

Conclusions

1. GenomicSuperSignature enables quick and easy access to the existing databases
2. GenomicSuperSignature is compatible to and, in some cases, outperform the existing models with the increased usability and scalability benefits
3. Biological signal learned from a specific dataset can be applied across different datasets, providing insight into weak or indirectly measured biological attributes

Overall, GenomicSuperSignature enables researchers to analyze new data in the context of existing databases, using minimal computing resources.



Availability

Package: https://shbrief.github.io/GenomicSuperSignature/
Demo: https://shbrief.github.io/GenomicSuperSignaturePaper/
Planned for release in Bioconductor 3.13

References

1. Greene, C. *at al.*, refine.bio: a resource of uniformly processed publicly available gene expression datasets.
2. Mao, W., Zaslavsky, E., Hartmann, B. M., Sealfon, S. C. & Chikina, M. Pathway-level information extractor (PLIER) for gene expression data. *Nat. Methods* 16, 607–610 (2019).
3. Ma, S. *et al.* Continuity of transcriptomes among colorectal cancer subtypes based on meta-analysis. *Genome Biol.* 19, 142 (2018).
4. Taroni, J. N. *et al.* MultiPLIER: A Transfer Learning Framework for Transcriptomics Reveals Systemic Features of Rare Disease. *Cell Syst* 8, 380–394.e4 (2019).
5. Becht, E. *et al.* Estimating the population abundance of tissue-infiltrating immune and stromal cell populations using gene expression. *Genome Biol.* **17**, 218 (2016)